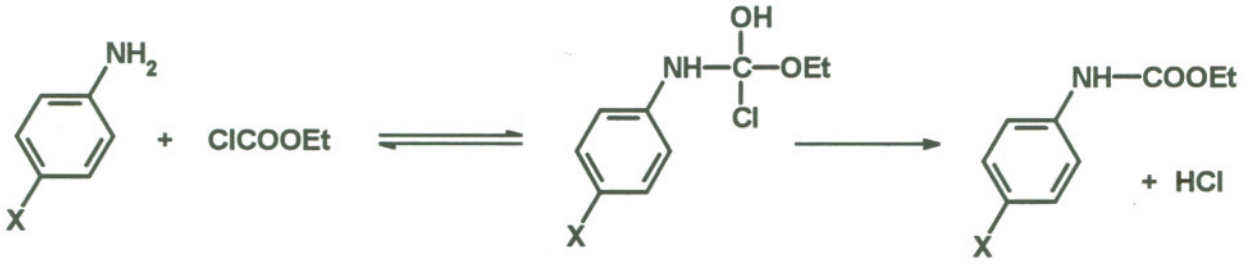
Organic Cume, February 10, 1996

Instructions: answer three out of four

1. Arrange the following compounds in order of solvolytic reactivity. Explain.

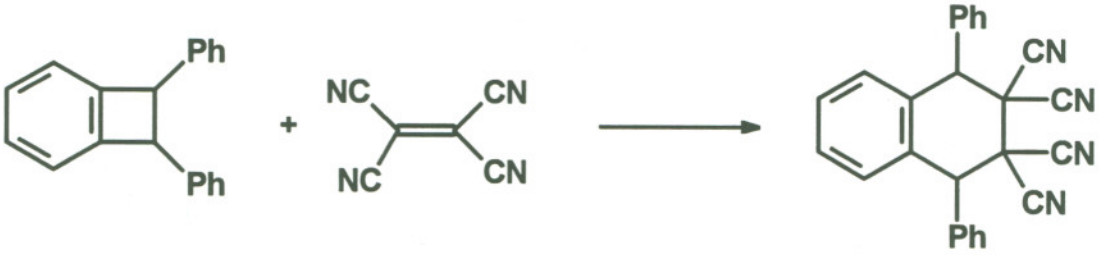
A B C D E

2. The reaction of anilines with ethyl chloroformate involves the two steps shown below. The Hammett plot shows two distinct linear regions: $\rho = -5.5$ for electron donating substituents and $\rho = -1.6$ for electron withdrawing substituents. Explain.



3. Which of the following two equilibria would you predict to lie farther to the right? Explain.

4. Suggest a reasonable mechanism for the following reaction if the observed rate law is first-order in compound **A** only.



A